

A Short Review of Simulation-Based Inference

Scientific Machine Learning Seminar

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Abstract

Simulation-Based Inference (SBI) offers a novel approach to solve inverse problems powered by machine learning. Hereby, a neural network learns to map observed data to a posterior distribution over model parameters. By using simulations, SBI does not require any likelihood evaluation which otherwise often is computationally costly or has to be approximated. This can allow SBI to outperform classical Bayesian inference methods in inference speed and posterior accuracy. However, common problems like model misspecification, neural inaccuracies or miscalibrated posteriors can limit the usefulness of SBI if they are not circumvented. This report presents basic SBI methodology, in particular Neural Posterior Estimation, discusses its advantages and limitations and reviews two exemplary applications for SBI: detecting planetary events from gravitational microlensing and inferring black hole merger parameters from gravitational waves.

1 Introduction

In computational physics there are usually two types of problems to solve: Firstly, forward problems where the parameters and/or boundary conditions of a model are given and the goal is to find data or a data distribution which adheres to this model. For example in the case of given parameters and initial values this is a typical simulation.

Though secondly, there are also inverse problems where the data is the preset and now the parameters or parameter distributions of the assumed model have to be inferred. So here the data likely comes from an observation and the parameters which represent physical properties of the system have to be computed. This is the type of problem that will be explored in this report.

A rather simple case of solving an inverse problem is the use of linear regression to fit a function to a given dataset. But for more complex models also methods with higher efficiency are needed. The classical approach here is to use Bayesian methods to infer the parameter posterior distribution, but Simulation-Based Inference introduces machine learning techniques to solving inverse problems. SBI is able to solve these problems in some cases more efficiently and computationally cheaper. Applications where SBI shows advantages are (but not limited to) gravitational microlensing (Zhang et al., 2021), gravitational wave physics (Dax et al., 2023) and neuroscience (Deistler et al., 2025, Sec. 4.2).

In Section 2 the general concepts of classical Bayesian inference are first explored, specifically focussing on Markov-Chain-Monte-Carlo methods. Then SBI and its underlying concepts are

introduced in Section 3 and exemplified in Section 4 with the help of a toy-example. The advantages and limitations of SBI are presented in Section 5. Two exemplary applications with results are portrayed in Section 6 and a conclusion is given in Section 7.

2 Classical Bayesian Inference

The most outstanding feature of Bayesian Inference in comparison to regression is that the output of BI is a posterior distribution of the parameters instead of just the most likely parameter set sampled from this distribution. So instead of just computing $\theta \sim p(\theta|x)$, the result is the full $p(\theta|x)$ distribution. The naming convention here is that θ is the set of parameters from the model and x is the observed data.

The underlying concept is the Bayes equation

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)}p(\theta) \quad (1)$$

from Bayesian statistics where $p(\theta|x)$ is the posterior distribution, $p(x|\theta)$ the likelihood, $p(x)$ the evidence and $p(\theta)$ the prior distribution. While the prior can be chosen adequately according to assumed distributions of the model parameters and the evidence is just a normalizing factor, the likelihood $p(x|\theta)$ has to somehow be calculated, evaluated or approximated in numerical methods of inferencing the posterior. This is the most important and also most costly step of classical Bayesian Inference (Deistler et al., 2025).

Different methods of Bayesian inference solve this in different ways. Markov chain Monte Carlo (MCMC) methods are relatively simple and widely used for inferring posteriors. These evaluate the likelihood many times randomly in the parameters space and build a posterior distribution from that. In Metropolis-Hastings, a MCMC method, the likelihood value at a random point in the parameter space determines if this point is accepted or rejected. The accumulated distribution of the accepted points then converges to the posterior after many evaluations (Prado et al., 2026).

There are adjusted methods that can make the algorithm more efficient (like introducing random walk instead of true random sampling) but all classical Bayesian inference methods suffer from similar key limitations: The convergence to a posterior that is accurate enough for the specific application takes many evaluations of the likelihood and is therefore very computationally expensive. For the applications that are later shown in Section 6 this can be up to 1 second per evaluation, for an adequately sampled posterior distribution the computation time per analyzed dataset can be huge.

In complex models the exact evaluation of the likelihood is often not possible or a lot more computationally expensive, so it has to be approximated by a Gaussian distribution, usually as part of the χ^2 method. This inserts another error source into the posterior distribution.

Classical Bayesian inference methods also have the problem that they cannot model multimodal posteriors very well. The algorithm often gets stuck in one mode and is unable to jump to another, this becomes a problem in models with a lot of close degeneracies (Deistler et al., 2025). In turn, grid searches and human intervention are often needed which increases the time and effort of the data analysis process. This may not be a possibility for the huge amounts of data that needs to be processed in modern science.

3 Introduction to Simulation-Based Inference

The key feature of Simulation-Based Inference is the skipping of the computationally expensive likelihood evaluation. This is the reason SBI is sometimes also called Likelihood-Free Inference. Instead of explicitly using a method to evaluate the likelihood at specific points in parameter space, the likelihood is implicitly given when sampling data from a simulator of the model. A neural network will train on this simulated dataset to infer a parameter posterior distribution from an observation. The likelihood still exists inside the dataset but does not have to be calculated or approximated.

This report focusses on the direct inference of posteriors, this methodology is called Neural Posterior Estimation (NPE). This is in contrast to Neural Likelihood Estimation (NLE) and Neural (Likelihood-)Ratio Estimation (NRE), they respectively require an additional step of computation to achieve the posterior but may offer advantages or more insight in the analysis process.

The SBI workflow can be subdivided into three steps as shown in Fig. 1: First, the simulator has to generate a relatively large dataset in accordance with the physical model. To achieve this, an adequate parameter prior distribution is chosen according to guesses on the physical properties of the model, e.g. the range of specific parameters. The simulator then samples x_i from the likelihood $p(x|\theta_i)$ and therefore generates a (x_i, θ_i) dataset. The likelihood is implicitly contained in the dataset but because the simulation solves a forward problem, this step is not very computationally costly.

Next, a neural network is trained to map each tuple (x_i, θ_i) to an estimated posterior density $q_\phi(\theta_i|x_i)$. The network architecture can in theory be chosen freely but Normalizing Flows is commonly used. The loss function

$$L(\phi) = E_{()} \quad (2)$$

is minimized to improve the network weights. It consists of the expectation value over all sampled tuples of the negative logarithm of the estimated posterior. This is very similar to the definition of a cross entropy and will result in a balancing act between low loss values and high confidence in the posterior. Maximizing $q_\phi(\theta_i|x_i)$ at the correct θ_i has the drawback of having to be more confident by making the distribution more slim. So the estimated posterior density $q_\phi(\theta_i|x_i)$ converges to the true posterior $p(\theta_i|x_i)$.

Lastly, the observed datapoint x_{obs} is inserted into the network and evaluated inside an interval of θ at any resolution. This will output the posterior distribution of any observation with relatively low computational cost. It is worthy to note that there is no need for further training when analyzing new data under the assumption of the same model (Deistler et al., 2025).

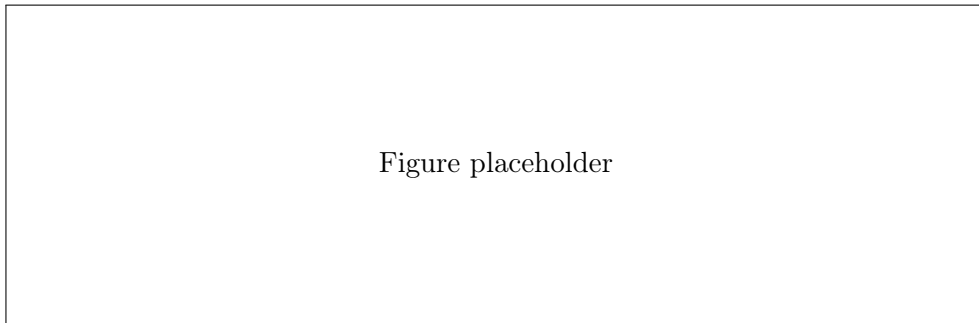


Figure 1: Workflow of SBI.

4 Simple SBI Example

A fairly simple demonstration of SBI is done in the following section with the help of the simulation of an angled throw. The structure and code of this example is adapted from [Deistler et al. \(2025\)](#). A ball is thrown from the origin at a specific angle θ . This is the sole parameter of the model. The ball lands on the ground at a specific distance from the origin, again this is the only data value x . The system behaves according to the physical laws of gravity, the flight trajectory should be a parabola. To turn the model probabilistic and make it harder for SBI to infer the posterior, a varying wind speed is added as a noise source.

Now it is not trivial to infer the posterior distribution of throwing angles from just a single distance measurement. Even Classical Bayesian methods would have problems since the posterior has two modes, a short throw could have come from either a very low angle or very high angle. A few example trajectories are shown in Fig. 2a.

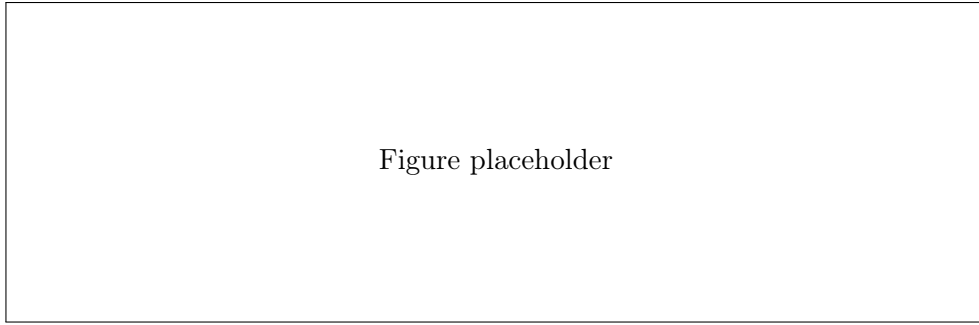


Figure 2: Workflow of SBI.

A parameter prior for throwing angle θ also needs to be chosen, here a Gaussian distribution which is limited between 0° and 90° and has its maximum at 45° (as shown in Fig. 2b) is a logical choice. Parameter values θ_i are sampled from this prior and put into the simulator that contains the probabilistic model of the angled throw as described above. A lot of angle - distance tuples (θ_i, x_i) are created, building the dataset that is shown in Fig. 2c.

This dataset is used to train a neural network based on the architecture of Normalizing Flows. The input of the network will be a tuple (θ_i, x_i) while the output is a predicted posterior density value q_ϕ at specifically $\theta = \theta_i$ in the parameter space under the assumption $x = x_i$. This is shown as a schematic in Fig. 2d. The network is optimized with the loss function in Eq. (2) as explained in Section 3.

With the trained network a posterior distribution is generated by inputting the observation x_{obs} , for example $x_{\text{obs}} = 10\text{m}$ and sampling the posterior in an interval of angles θ values. This is done for a few hypothetical observations and sampled for $0^\circ < \theta < 90^\circ$. The sampling rate of θ can be chosen freely but affects the computational cost of the algorithm and may not offer any advantages when increasing it above a neural estimation accuracy. Though in this very simple example the computational cost is negligible. The results are shown in Fig. 2e. For smaller x_{obs} the two modes in the posterior are easily visible as two separate peaks.

5 Benefits and Limitations

What SBI is able to improve upon compared to classical Bayesian Inference can be put into three main benefits. First of all, the inference of the posterior is likelihood-free. There is no

need to explicitly evaluate the likelihood at random points in the parameter space like it is the case for MCMC, the likelihood is implicitly contained inside the dataset from the simulator the neural network is trained on. That means the inference is a lot faster and there is no need for approximations of the likelihood in the process like it is often done for Bayesian Inference.

Adjacent to that, SBI also offers the concept of Amortized Inference. Here the largest amount of computational cost is paid upfront when running the simulator and training the neural network, but the actual inference by the network is very fast and cheap. This can speed up the analysis for research purposes a lot if a model is reused for a lot of data since there is no additional simulating and training needed after the weights for a network are optimized. For complex models this is especially relevant, in classical Bayesian Inference a considerable amount of computation time would be needed at each likelihood evaluation.

And thirdly, SBI is able to model the multimodality of posteriors very well. It does not have to explore different modes manually like it is often the case for MCMC where the algorithm can get stuck inside a specific mode. The neural network is aware of the entire posterior at once, not only dependent on a local likelihood approximation.

Though bundled with these advantages there are also limitations that exist in SBI algorithms. The previously mentioned upfront simulation and training due to Amortized Inference may be a drawback when there is only a limited amount of observations to be processed. Therefore, the gain in inference speed is may not be worth it.

Additionally, a common pitfall of SBI is model misspecification. When giving the neural network an observation that has not been simulated before because such a case is not included in the model, the network will output a nonsensical posterior distribution that is not related to real physics. To illustrate, when giving the trained network from the example in Section 4 a distance larger than any possible throw, the output will be an unpredictable posterior of angles that would never lead to such a distance. In complex models this is not trivial to avoid.

It is important to remember that the output posterior is only a neural estimation and its accuracy is dependent on the architecture of the neural network as well as training. There is no convergence guarantee analogous to MCMC to assure that the estimated posterior is getting closer to the true posterior. When a posterior does not meet the quality criteria it is not possible to run the algorithm for longer to attain a higher accuracy, only an entirely new simulation and training run can improve the quality of posterior.

A common problem of posteriors inferred via NPE is that they may be miscalibrated. They can have biases, meaning a shifted mean to either side, or are either over- or underconfident, resulting in a distribution that is spread out not enough over too much. Though miscalibration can be diagnosed quite effectively with the help of quantile-occurrence plots. The posterior probability of the true value is compared with how often it actually occurs in a validation dataset and can therefore show errors in posterior accuracy on a global scale. If miscalibration is hinted at in the quantile-occurrence plots, changes to the network and training are probably necessary (Deistler et al., 2025).

6 Application

The research area of gravitational microlensing is one of the prime examples where SBI shines in the data analysis pipeline. When a star with an exoplanet in orbit moves into the line of sight between earth and a bright background star the light of background star is lensed around

the foreground star and exoplanet and magnified in a very specific way. This effect cannot be spatially resolved but telescopes can measure the change of brightness as a lightcurve. The shape of the lightcurve is defined by the physical properties of both stars and exoplanet resulting in a large amount of parameters (around 11) and a complex model while the data is often limited. Additionally, the model is also highly degenerate and often has multiple complex modes that have to be explored manually when using classical Bayesian Inference.

This is the reason [Zhang et al. \(2021\)](#) is testing the capabilities of SBI for microlensing data from the Roman Space Telescope. The training was done with a dataset of more than 275,000 simulated events and about 25 hours. SBI was able to increase the computation speed drastically from 1 simulation per second which would be needed for one MCMC step to 10^5 posterior samples per second. What was found is that all the posterior modes were accurately modeled though some biases existed in the posterior. The authors recommend following up on SBI with MCMC to converge on an accurate posterior and also the most likely solution. This demonstrates SBI's functionality in the data analysis pipeline for quickly exploring multimodality of posteriors and attaining initial values for further inference methods.

Similarly, the models needed to analyze gravitational wave signals from black hole mergers are dependent on a lot of parameters but only very limited data is available. [Dax et al. \(2023\)](#) use SBI in their data analysis pipeline for gravitational wave events but implement Flow Matching Posterior Estimation (FMPE) in their SBI architecture. Instead of Normalizing Flows FMPE is based on a Flow Matching neural network. This learns a vector field transformation from a simple distribution to the real posterior and not the transformation of all the datapoints, achieving fewer parameters to describe the same transformation, a more expressive neural network and 30% reduced training time. In turn, the inference time is increased since the sampling also requires solving an orthogonal differential equation dependent on the velocity field output.

Generally all physics or science research fields which are heavy in data processing would likely benefit from an implementation of SBI in specific contexts. Other fields of research noted in [Deistler et al. \(2025\)](#) for SBI applications are psychophysics and neuroscience.

7 Conclusion

Benefits and Limitations (stuff from abstract)

When to use SBI inverse problem, time critical inference, expensive likelihood evaluation, multimodal posterior

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